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FIRST CLASS BIOMEDICAL SCIENCE GRADUATE | HEALTH DATA AND ANALYSIS | LABORATORY SCIENCE | HEALTHCARE INNOVATION

PROFILE

First Class Biomedical Science graduate with experience in governed health-data analysis, NHS-facing commercial research, laboratory practice and scientific writing. In the Genomics England Trusted Research Environment, I built and corrected a reproducible clinical machine-learning workflow for a Brugada-suspect cohort. At Health Innovation East, I turned clinical, market and policy evidence into business-case inputs, competitor work and practical guidance for innovators. I have since carried the same discipline into public MRI, evidence-readiness and facility-science projects, keeping the data, method and decision visible enough to check.

CORE CAPABILITIES

- **Health data and analytics:** Python workflows, linked longitudinal hospital data, cohort construction, controlled clinical vocabularies, statistical reasoning, machine learning, patient-separated evaluation and visible quality control.
- **Healthcare evidence and commercial research:** Evidence appraisal, market intelligence, competitor analysis, NHS pathway reasoning, business-case support, health-technology adoption and stakeholder-facing guidance.
- **Laboratory and biomedical science:** Molecular biology, microbiology, pharmacology, toxicology-data quality, experimental interpretation, contamination control and source-backed facility-science preparation.
- **Scientific communication and delivery:** Literature research, evidence briefs, technical documentation, dissertation and poster writing, presentations, project tracking, Git/GitHub and audience-aware explanation.

EDUCATION

BSc (Hons) Biomedical Science with Placement Year, Anglia Ruskin University, Cambridge

September 2022 - June 2026 | First Class Honours

- Undergraduate Project 82%, including an 83% dissertation; Mathematics for the Biosciences 91%; General Microbiology 84%.
- Further study covered genetics, molecular biology, diagnostic pathology, pharmacology, translational medicine, statistics, research design and ethics.

PROFESSIONAL EXPERIENCE

Commercial Intern, Health Innovation East, Cambridge

July 2024 - July 2025

- **AI adoption guidance:** Authored an AI Innovators' Toolkit that translated MHRA/UKCA, NICE, DTAC, data-governance, interoperability and risk requirements into practical guidance for health-technology teams.
- **Market intelligence:** Took ownership of learning CB Insights and turned it into dashboards, live databases and repeatable horizon-scanning resources, contributing to evidence coverage across approximately 50 technologies.
- **NHS adoption and business cases:** Supported pathway, evidence-pack and business-case work for a Parkinson's cueing device, joining clinical evidence, patient benefit, NHS priorities and cost assumptions for pilot discussions.
- **Commercial and policy analysis:** Mapped an AI-enabled ECG technology against competing approaches and 2025/26 QOF incentives, then tested the adoption reasoning with a GP, health economist and pathway specialist.
- **Stakeholder delivery:** Contributed to innovator calls, presentations and Integrated Care Board workstream mapping while managing toolkit, competitor and horizon-scanning requests in parallel.

Research Intern, ConsoneAI / DioScor and Anglia Ruskin University, Cambridge

May 2024 - June 2024

- Mapped ECHA and PubChem toxicology fields to the DioScor platform, creating a matching matrix for consistent downstream use of dose-toxicity evidence.
- Wrote Python scripts to extract, collate and clean data, then structured the outputs in Excel and CSV for analysis and platform population.
- Investigated differences in terminology, administration route, endpoint and organism metadata before integration, supported by targeted literature review and 3Rs reasoning.
- Presented the method and findings through a poster at the ARU Postgraduate Research Conference 2024.

SELECTED PROJECT PORTFOLIO

Genomics-linked Brugada syndrome analysis, Genomics England Trusted Research Environment

Undergraduate Project | September 2025 - May 2026

- Built and documented a multi-stage Python workflow around a consent-filtered Brugada-suspect cohort in a governed research environment.
- Linked sequencing dates to admitted-patient, outpatient and emergency-care records, applied pre-sequencing chronology controls and engineered participant-level ICD-10, OPCS-4 and healthcare-utilisation features.
- Developed configuration-driven feature and target definitions, validation checks, row-count reconciliation, run manifests and decision logs so analytical choices remained reviewable.
- Compared logistic-regression, random-forest and neural-network baselines using split-first preprocessing, repeated evaluation and class-imbalance-aware metrics.
- Corrected full-cohort preprocessing after identifying a leakage risk. Repeated runs showed modest, seed-sensitive performance, leading me to close the model branch and define the next evidence needs.

Promoter-linked clinical signal audit

Post-dissertation research extension | July 2026 - August 2026

- Designed and ran a staged audit of whether five years of pre-sequencing hospital records held reproducible information about promoter genotype, using predeclared gates, permutation controls and an external-covariate audit.
- Phase 1 found no features surviving family-wise error control at 0.05; Phase 2 binary representations produced negative mean log-loss gain against prevalence.
- Closed the model search under the written stop/go rules and retained the covariate lineage, phase reports and decision record for future research design.

Public multiphase liver MRI analysis

Independent project | August 2026

- Audited a public 17-patient, four-phase MRI dataset, including NIfTI geometry, phase provenance, supplied co-registration and two-rater liver and tumour masks.
- Fitted a within-patient intensity scale, constructed explicit enhancement features and evaluated a voxel-level tumour-probability baseline after separating patients.
- Ran a four-channel 2D U-Net smoke test under the same patient split through a seeded bash workflow with environment, checksum and output gates, then reviewed its errors alongside the transparent DCE baseline.

Clinical informatics trial-operations demonstration

Self-directed project | June 2026 - present

- Built a deterministic synthetic multicentre Phase II metabolic study with eight linked tables covering sites, participants, visits, laboratories, adverse events, queries, protocol deviations and milestones.
- Translated protocol requirements into relational fields, relationships and validation rules before producing study and site summaries.
- Created record-level checks for consent, eligibility, randomisation, visit windows, laboratory completeness, unresolved-query age, adverse-event follow-up and milestone timing.
- Built Plotly and Streamlit review views whose filters recalculate from the retained records, supported by pytest checks and GitHub Actions. Repository: github.com/JJZirebwa/clinical-informatics.

Rare-disease evidence readiness and phenotyping methods

Independent methods research | May 2026 - present

- **Evidence readiness and measurement:** Developed a structured way to test whether available evidence can observe an intended biological or clinical target before modelling begins.
- **Provenance-aware phenotyping:** Designed a method-development framework for biological priors, feature provenance, anti-circularity controls and measurement prioritisation in weak-signal rare-disease work.
- Uses explicit checks for ascertainment, leakage, unstable signal, source dependence and the next measurement or linkage required by the question.
- Applies the completed Brugada work as an initial case and keeps evidence sources, assumptions, method changes and validation questions linked for later review.

LABORATORY AND SCIENTIFIC PRACTICE

DNA manipulation and protein-expression practical sequence

University laboratory programme

- Completed PCR amplification, agarose gel electrophoresis, purification and restriction digestion, ligation, bacterial transformation, colony PCR, plasmid mini-prep and restriction mapping across a staged cloning workflow.
- Used RFLP analysis for SNP discrimination and SDS-PAGE with Coomassie staining to assess protein-expression patterns, supported by primer design, sequence alignment, lab calculations and precision pipetting.

Microbiology identification and antimicrobial-testing investigation

Assessed laboratory practical | Practical mark 90%

- Used aseptic technique, agar preparation, streak and spread plating, Gram staining, oxidase and catalase tests, CLED interpretation and a dichotomous key to identify an unknown organism.
- Performed antimicrobial disk diffusion, colony counting and CFU calculations, then combined the test results into a reasoned identification.

Pharmacology modelling and dose-response analysis

Assessed scientific project

- Modelled TRPV6 receptor structure and used primary literature to interpret selectivity, inhibition and disease relevance through annotated figures.
- Ran a virtual organ-bath analysis, generated dose-response curves, calculated ED50 values and interpreted shifts produced by agonists, antagonists and enzyme inhibition.
- **Laboratory quality:** Contamination control, PPE and safe sample handling, protocol discipline, result interpretation and structured documentation; academic grounding in SOPs, LIMS, IQC/EQA, UKAS and ISO concepts.

TECHNICAL TOOLKIT

- **Programming and analysis:** Python, pandas, NumPy, scikit-learn, matplotlib, logistic regression, random forests, neural-network baselines, feature engineering, statistical visualisation and Excel/CSV workflows.
- **Clinical and genomic data:** SQL-style LabKey querying, linked longitudinal records, ICD-10, OPCS-4, data dictionaries, cohort logic, focused VCF inspection with bcftools and sample-to-participant mapping.
- **Medical imaging:** NIfTI geometry quality control, SimpleITK registration inspection, multiphase DCE feature construction, voxel probability maps, PyTorch and a small four-channel 2D U-Net baseline.
- **Reproducible delivery:** Bash, Git/GitHub, configuration controls, tests, checksums, run manifests, audit logs, decision records, Microsoft Word, Excel and PowerPoint, and CB Insights.
- **Developing tools:** Introductory R from bioscience statistics teaching and continued practice through analytical preparation.

ADDITIONAL PROJECTS AND WRITING

Facility-science readiness for proteomics, metabolomics and LC-MS workflows

Independent project | July 2026 - present

- Built a source-backed preparation workspace covering sample handover, preparation risks, QC controls, metadata, data interpretation, FAIR practice and sustainable facility work.
- Created a Python sample-sheet validator for required metadata, duplicate IDs, batch and injection-order collisions, and minimum QC controls.
- Created a source-index builder that converts a reviewed Markdown source register into JSON and CSV for traceable reuse.

Scientific and health writing

- Produced an 83% dissertation, a conference research poster, evidence briefs, technical summaries, slide decks and practical guidance for scientific, clinical and innovation audiences.
- Wrote a long-form narrative-medicine essay examining sickness, embodiment and well-being through lived experience and biomedical training.

LEADERSHIP AND ADDITIONAL INFORMATION

- Secretary, Cambridge Early Learning Innovation Network.
- English and Shona, fluent; French, beginner.
- Earlier education at St George's College, Harare included A Levels in Biology, Chemistry and Mathematics, with AS English (2021).
- Based in Cambridge, United Kingdom.